

k8s Task04 Resource Quota/limit Range

1. Create namespace test

```
# mkdir 04.k8s-task
# cd 04.k8s-task
# kubectl create ns test
# kubectl get ns
```

```
root@k8s-control:~# mkdir 04.k8s-task
root@k8s-control:~# cd 04.k8s-task/
root@k8s-control:~/04.k8s-task# kubectl create ns test
namespace/test created
root@k8s-control:~/04.k8s-task# kubectl get ns
NAME                STATUS    AGE
default             Active   19d
kube-node-lease     Active   19d
kube-public         Active   19d
kube-system         Active   19d
test                Active    8s
root@k8s-control:~/04.k8s-task#
```

2. Create pod nginx

```
# vim pod-nginx.yaml
```

```
apiVersion: v1
kind: Pod
metadata:
  name: nginx
  namespace: test
spec:
  containers:
  - name: nginx
    image: nginx:latest
    resources:
      requests:
        memory: "2Gi"
        cpu: "500m"
      limits:
        memory: "2Gi"
        cpu: "500m"
```

```

apiVersion: v1
kind: Pod
metadata:
  name: nginx
  namespace: test
spec:
  containers:
  - name: nginx
    image: nginx:latest
    resources:
      requests:
        memory: "5Gi"
        cpu: "500m"
      limits:
        memory: "5Gi"
        cpu: "500m"

```

kubectl apply -f pod-nginx.yaml

kubectl get pods -n test

```

root@k8s-control:~/04.k8s-task# kubectl apply -f pod-nginx.yaml
pod/nginx created
root@k8s-control:~/04.k8s-task# kubectl get pods -n test
NAME      READY   STATUS    RESTARTS   AGE
nginx     1/1     Running   0           25s
root@k8s-control:~/04.k8s-task# █

```

3. Create ResourceQuota

vim quota.yaml

apiVersion: v1

kind: ResourceQuota

metadata:

name: pod-limit-and-quota

namespace: test

spec:

hard:

requests.cpu: "500m"

requests.memory: "5Gi"

limits.cpu: "500m"

limits.memory: "5Gi"

```

apiVersion: v1
kind: ResourceQuota
metadata:
  name: pod-limit-and-quota
  namespace: test
spec:
  hard:
    requests.cpu: "500m"
    requests.memory: "5Gi"
    limits.cpu: "500m"
    limits.memory: "5Gi"

```

kubectl apply -f quota.yaml

kubectl describe quota pod-limit-and-quota -n test

```

root@k8s-control:~/04.k8s-task# kubectl apply -f quota.yaml
resourcequota/pod-limit-and-quota created
root@k8s-control:~/04.k8s-task# kubectl describe quota pod-limit-and-quota -n test
Name:                pod-limit-and-quota
Namespace:            test
Resource              Used    Hard
-----
limits.cpu            500m   500m
limits.memory         5Gi    5Gi
requests.cpu          500m   500m
requests.memory       5Gi    5Gi
root@k8s-control:~/04.k8s-task#

```

4. Create pod nginx2

vim pod-nginx2.yaml

```
apiVersion: v1
```

```
kind: Pod
```

```
metadata:
```

```
  name: nginx-2
```

```
  namespace: test
```

```
spec:
```

```
  containers:
```

```
    - name: nginx
```

```
      image: nginx:latest
```

```
      resources:
```

```
        requests:
```

```
          memory: "2Gi"
```

```
          cpu: "500m"
```

```
        limits:
```

```
          memory: "2Gi"
```

```
          cpu: "500m"
```

```

apiVersion: v1
kind: Pod
metadata:
  name: nginx-2
  namespace: test
spec:
  containers:
    - name: nginx
      image: nginx:latest
      resources:
        requests:
          memory: "2Gi"
          cpu: "500m"
        limits:
          memory: "2Gi"
          cpu: "500m"

```

kubectl apply -f pod-nginx2.yaml

```

root@k8s-control:~/04.k8s-task# kubectl apply -f pod-nginx2.yaml
Error from server (Forbidden): error when creating "pod-nginx2.yaml": pods "nginx-2" is forbidden: exceeded quota: pod-limit-and-quota, requested: limits.cpu=500m,limits.memory=2Gi,requests.cpu=500m,requests.memory=2Gi, used: limits.cpu=500m,limits.memory=5Gi,requests.cpu=500m,requests.memory=5Gi, limited: limits.cpu=500m,limits.memory=5Gi,requests.cpu=500m,requests.memory=5Gi
root@k8s-control:~/04.k8s-task# █

```

5. Create namespace limit

kubectl create ns limit

6. Create LimitRange

vim limitrange.yaml

```

apiVersion: v1
kind: LimitRange
metadata:
  name: cpu-resource-constraint
  namespace: limit
spec:
  limits:
    - default:
        memory: "500m"
      defaultRequest:
        memory: "500m"
      max:
        cpu: "1"
      min:
        cpu: "100m"
      type: Container

```

```

apiVersion: v1
kind: LimitRange
metadata:
  name: cpu-resource-constraint
  namespace: limit
spec:
  limits:
    - default:
        memory: "500m"
      defaultRequest:
        memory: "500m"
      max:
        cpu: "1"
      min:
        cpu: "100m"
      type: Container

```

```
# kubectl apply -f limitrange.yaml
```

```
# kubectl describe limitranges cpu-resource-constraint -n limit
```

```

root@k8s-control:~# kubectl create ns limit
namespace/limit created
root@k8s-control:~# kubectl apply -f limitrange.yaml
limitrange/cpu-resource-constraint created
root@k8s-control:~# kubectl describe limitranges cpu-resource-constraint -n limit
Name:      cpu-resource-constraint
Namespace: limit
Type:
Resource  Min  Max  Default Request  Default Limit  Max Limit/Request Ratio
-----
Container  cpu   100m  1    1    500m             1              -
Container  memory -     -    500m             500m           -
root@k8s-control:~# █

```

7. Deploy another pod with conflicting resource requests

```
# vim pause-pod.yaml
```

```
apiVersion: v1
```

```
kind: Pod
```

```
metadata:
```

```
  name: pause-pod
```

```
  namespace: limit
```

```
spec:
```

```
  containers:
```

```
    - name: pause
```

```
      image: registry.k8s.io/pause:2.0
```

```
      resources:
```

```
        requests:
```

```
          cpu: 700m
```

```

apiVersion: v1
kind: Pod
metadata:
  name: pause-pod
  namespace: limit
spec:
  containers:
  - name: pause
    image: registry.k8s.io/pause:2.0
    resources:
      requests:
        cpu: 700m

```

kubectl apply -f pause-pod.yaml

```

root@k8s-control:~# kubectl apply -f pause-pod.yaml
Warning: spec.containers[0].resources.limits[memory]: fractional byte value "500m" is invalid, must be an integer
Warning: spec.containers[0].resources.requests[memory]: fractional byte value "500m" is invalid, must be an integer
pod/pause-pod created
root@k8s-control:~#

```

8. Edit pause-pod.yaml

vim pause-pod.yaml

```

apiVersion: v1
kind: Pod
metadata:
  name: pause-pod
  namespace: limit
spec:
  containers:
  - name: pause
    image: registry.k8s.io/pause:2.0
    resources:
      requests:
        cpu: 700m
      limits:
        cpu: 700m

```

kubectl apply -f pause-pod.yaml

```

root@k8s-control:~# kubectl apply -f pause-pod.yaml
Warning: spec.containers[0].resources.limits[memory]: fractional byte value "500m" is invalid, must be an integer
Warning: spec.containers[0].resources.requests[memory]: fractional byte value "500m" is invalid, must be an integer
pod/pause-pod created
root@k8s-control:~# kubectl apply -f pause-pod.yaml
pod/pause-pod unchanged
root@k8s-control:~#

```